Pharma Market Sizing & Drug Launch Strategy

Executive Summary - illustrative US migraine commercial model

Fictional case: Calystren Therapeutics and Project CYR-217 are hypothetical. The model represents a post-approval commercial launch case. Public evidence, analyst assumptions, and hypothetical client inputs are kept separate.

Commercially Addressable Pool
1.15M
modeled patients

Base Year 5 Share
12%
illustrative scenario assumption

Base Net Price
$7,500
derived: $12,500 WAC × (1 – 40% GTN)

Base Year 5 Modeled Annual Net Sales
$1.038B
modeled annual net sales

Patient funnel	Modeled patients
US adults 18+	**262,083,034**
Broad migraine/severe-headache proxy	**40,098,704**
Diagnosed population	**22,535,472**
Modeled treatment-engaged population	**7,211,351**
Preventive candidate pool	**2,884,540**
Commercially addressable pool	**1,153,816**

Base revenue model

Year 1: $130M
Year 2: $346M
Year 3: $649M
Year 4: $865M
Year 5: $1.038B
5-year cumulative: $3.029B

Revenue Sensitivity Analysis

Market-share range (8%–16%): $692M spread
Net-price range ($6k–$9k): $415M spread
Addressable-pool range (±20%): $415M spread

Key findings

1. The modeled funnel produces a **1.15M** commercially addressable patient pool.
2. At **12% Year 5 share**, the model reaches about **138k** modeled product patients and **$1.038B** Year 5 modeled annual net sales.
3. Within the selected ranges, the largest Year 5 revenue spread is from **market share (~$692M)**; net price and patient-pool spreads are each about **$415M**.

Risks / limitations

1. The patient funnel combines evidence from studies with different populations, years, and definitions; the result is illustrative rather than claims-based.
2. Treatment engagement, commercial access, market share, pricing, and scenario ramps are modeling assumptions, not observed launch outcomes.
3. Competitor sales are reported at different geographic/indication scopes, so they are factual context rather than directly comparable market shares.

Recommendation: Advance to the next stage of commercial validation. The modeled opportunity is meaningful enough to justify further work, subject to validating the patient pool, commercial access, achievable market share, pricing, and product positioning with better evidence.

Key evidence: US Census 2023; Burch et al. PMID 29527677; AMPP PMID 17371352; AHS 2024 DOI 10.1111/head.14692; Hepp et al. PMID 27837173.
Modeling note: WAC and GTN are inputs; net price is derived. The Illustrative Commercial Access Factor is commercial reachability, not a clinical requirement. Dollar opportunity is theoretical patient-pool value at the modeled net price.